**MIXDES 2007****Ciechocinek, POLAND**
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TOWARDS A 3D HETEROGENEOUS FUNCTIONAL INTEGRATION****J-L. SANCHEZ, A. BOURENNANE, M. BREIL, P. AUSTIN, M. BRUNET,
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ABSTRACT: This paper presents a brief overview of the monolithic integration in the field of power electronics. Emphasis is mainly put on the functional integration concept. The role that this mode of integration, according to its classical definition, played to enable the monolithic integration of the power device with auxiliary elements (mainly protections and supply) for the realization of new functions dedicated for medium power applications is highlighted. At that end, some of the recent realizations are described in order to showcase some of the potentialities of this mode of integration. Furthermore, to extend further the classical integration towards a **3D “heterogeneous” functional integration**, an example that highlights the improvements that should be achieved at the device’s level as well as at the device’s environment level, for the development of new power integrated functions for ac applications, is discussed. The last part deals with the technology process evolution for the realization of the active devices as well as the passive elements. In this part, a flexible technological process and its importance in the development of more complex functions, implemented in 3D within the silicon die volume and at the surface, is described in more detail.

INTRODUCTION

During the last years, an impressive progress has occurred in the field of power integrated circuits. This is, on one hand, due to the joint evolution of CMOS, BICMOS integrated circuit technologies and the lateral power MOS semiconductor devices (LDMOS) technologies and on the other hand due to the evolution of the isolation techniques.

Compared to signal processing integrated devices, the electrical problems involved are different in power electronic integration. Indeed, when designing power integrated circuits, one has to take into account high current and voltage levels, dV/dt and di/dt , thermal effects... However, smart power devices or high voltage integrated circuits mixing CMOS or BiCMOS technology, for control and processing circuits, and DMOS technology, for power devices, are still limited to low and medium voltage applications.

In the industrial sector (static supply, variable speed, ...) and more generally in the public at large (household appliances, small electric equipment, heating ...), numerous power conversion and protection applications connected to the electrical supply network can benefit from the advantages brought about by integration. Indeed, these technologies are employed, nowadays, to achieve integrated drivers for VDMOS or IGBT power devices. Even though, the use of these technologies enabled the integration of various devices on the same chip, some other design aspects still need to be addressed, as it will be highlighted later. In these applications, the range of voltage required (in excess of 600 V) leads to a vertical design of power devices. Furthermore, with respect to the current levels required (greater than ten amps), the power device uses the greater part of the integrated device area. So, the active area of this power device becomes larger

than signal processing circuits. To cut down on fabrication costs, the basic structure and technology used to obtain the integrated function have to be that of the power device and additional elements can be integrated by adding a minimum number of technological steps. The requirements of this type of integration ideally suits the capabilities of the « Functional Integration » concept already used in the field of power devices. Indeed, this mode of integration which combines M.O.S effect at the top side with the bipolar interactions occurring in the semiconductor volume is well suited to develop specific power devices and/or to integrate protection and control functions.

A logical evolution of the well known « classical functional integration » would be towards the « heterogeneous 3D functional integration ». This latter should allow the integration of diverse functions (control, protection, integrated gate driver power supply, isolation, storage, ...) with the power device. To that end, micro-fabrication techniques (materials deposition (other than silicon), etching, thick resists, wafer bonding, stacking, micro-assembly) must be employed. In order to cut-down on fabrication costs, the different technological steps should be based on techniques and equipments that make it possible to work at the silicon wafer level. Therefore, the association of materials (magnetic, piezoelectric, dielectric, ...) with silicon and the exploitation of silicon volume should be the main techniques to be used.

The trends then would be towards the integration of power functions either in a package or on the same chip. The 3D heterogeneous functional integration concept should make it possible to integrate passive and active elements. In the following, we will describe briefly the achievements made in the integration of power functions (smart power, classical functional integration) and the technology evolution for the fabrication of passive

elements (isolation, storage). We describe a sample technology process “a flexible technological process” that could make the 3D heterogeneous functional integration concept feasible.

BRIEF OVERVIEW OF THE POWER INTEGRATED FUNCTIONS

Historically, the strategy adopted for monolithically integrating new power functions is either based on enhancing the functionality (signal processing part) at the expense of the power devices performance or by favouring the power part [1, 2]. Indeed, The family of ‘smart power devices’ and ‘High Voltage Integrated Circuits (HVIC)’ belongs to the first category and are realized using the technology of integrated circuits (CMOS, BiCMOS), on the other hand the family of devices based on the functional integration concept proceeds from the second approach and are realized using the power device technology process.

a - *Power integrated circuits*: This type of integration, as illustrated by the block diagram given in figure 1, includes many parts among which we can state: power elements, control and processing circuits as well as interface part.

The control and processing part corresponds to a logic circuit achieved with CMOS with respect to its low power consumption and high density of integration. This part may contain microprocessors and digital signal processors (DSP). The power elements are generally lateral (LDMOS) or vertical (VDMOS). At the interface part, however, the trends consists in replacing bipolar circuits by BiCMOS circuits presenting better performances. It should be note that BCD technologies (using bipolar, CMOS and DMOS processes) enable greater flexibility for the realisation of this type of power integrated circuits.

The isolation between the power part and the low voltage part is achieved either by junction isolation or dielectric isolation. Since the choice of the type of isolation is mostly linked to cost, the junction isolation technique is generally the preferred solution for standard applications and the dielectric isolation is reserved for specific applications requiring a high level of isolation.

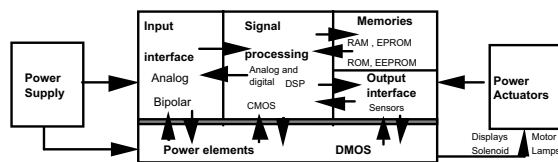


Figure 1: Block diagram of a « Smart power » device

The design rules used for the design and realisation of smart power devices derive from the design rules and technology of VLSI and ULSI devices. However, generally, the application of ULSI devices design rules to smart power devices takes place after some time delay. Indeed, according to the curves given in figure 2, in the early eighties the first smart power devices were realised using 2.5 or 4 μm design rules while the ULSI devices were realised using 1 μm

ones. It's only when submicron design rules were used for the realisation of ULSI devices that 1.5 or 2 μm were applied for the realisation of smart power devices.

This difference was essentially linked (i) to the more complex fabrication that must be taken into account: isolation, edge terminations for power devices, combination of different kinds of devices, CMOS, DMOS, bipolar, and (ii) to the important development of VLSI and ULSI devices driven by an important market. The technology developed for ULSI and VLSI circuits has always been used a few years later for Smart Power devices [3].

The major fields of application of these power integrated circuits are portable equipment, domestic appliance, lighting and the automotive industry. The development of power integrated circuits has been essentially driven by the automotive electronic requirement for applications such as motor control, fuel injectors drivers ...

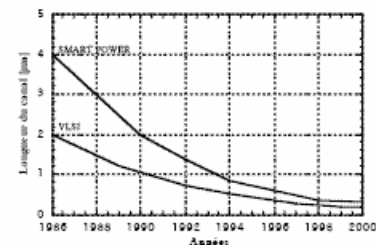


Figure 2. Evolution of minimum features in ICs and in smart power devices

b - *Functional integration*: according to this concept, the function results from the interaction between semiconductor layers judiciously arranged and sized and not from the interconnection of separate devices. The very first semiconductor structures dedicated to power, mainly thyristors and triacs correspond to the functional integration concept and more precisely they belong to the family of “bipolar-bipolar” type. The application of MOS technologies to power devices led to a new dimension of this concept by offering an additional degree of freedom provided by MOS effect at the surface of that can be combined with the bipolar interactions occurring in the semiconductor volume. Since then, new families of power devices have been developed combining the high current capability of bipolar devices with the simple drive capability of the MOS transistors. The concept of functional integration allowed to achieve new switching functions monolithically integrating the power device with the protection, amplification and validation of the control [4,5].

Discussion: according to this brief description, the technology process that is used for the realisation of a monolithically integrated function is either that of the signal processing part or that of the power device. However, the performance of the final integrated circuit does not take benefit of the last advances achieved in each technology. Indeed, as described previously and according to the curves (figure 2), on one hand, in smart power the design rules employed are at least “one generation” lagged behind those employed in ULSI

devices. Hence, due to the monolithic integration, the performance of the signal processing part integrated with the power part is decreased. Moreover, in some applications, this shortcoming might be unacceptable and another integration solution tacking into account new *functions partitioning* should be considered. To highlight this point, we rely on examples that are recently reported in literature.

PARTITIONING OF THE POWER INTEGRATED FUNCTIONS

In the field of power systems, the integration might involve both low voltage/power circuits and high voltage/power circuits. To achieve such integration, one can use technologies that permit to deal with both, low and high voltage/power circuits rather than technologies that can deal with only one circuit type. The big advantage of utilizing a technology that makes possible the integration of both circuit types is the cost but this is achieved at the price of a poorer performance of the integrated function. The use of two separate technologies that can deal with two different circuit types can allow putting side by side the power part and the signal processing part. With this latter, there is still a gain in terms of cost of the entire solution but the most important gain is related to the complex functions that can be realized in each circuit type (mainly in the signal processing part). Indeed, the possibility to benefit from the last advances reached in each technology allow to get the best performances. The different cases are illustrated hereafter through some examples.

In traditional Power Integrated Circuits (PICs), the Bipolar CMOS DMOS (BCD) technology process is generally used and it is sufficient for many applications. However, with the advancements in power device technologies, the power handling capability of the BCD process is not as sufficient as the new power devices when dealing with larger currents and voltages or when the space and heat sinking is minimized. But the drawback of the new power device technologies is that only limited signal processing functions can be economically added [6]. In applications, where fairly complex signal processing functions and lower power handling capability are required or where the thermal path is such that it can keep the device cool regardless in the generated power, the BCD process is used as illustrated in figure 3 [6]. In applications where devices are mainly handling power with limited signal processing, the power process alone will often work. An example, as reported in literature, is given in figure 4 that shows simple function such as drain to source voltage feedback signal and a gate drive disable control function [6].

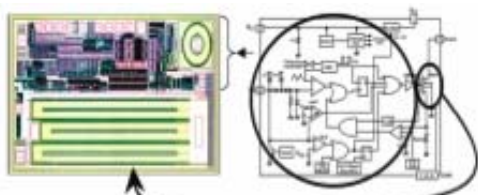


Figure.3. Single chip 700 V senseFET power switch in a 7 lead DIP; example of functions integration using BCD process [6].

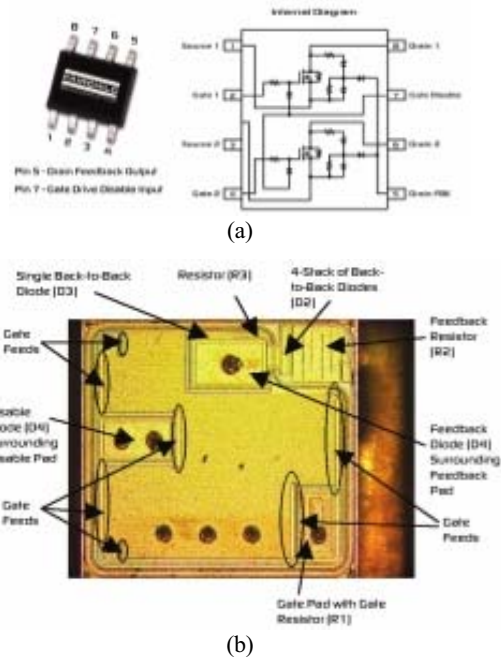


Figure. 4 a) Logic Level Gate Dual MOSFET in SO8 with a 5V logic level feedback signal
b) Die picture of the logic level gate dual MOSFET [6].

In the case that the integrated function requires complex signal processing functions and dedicated to high power applications then it is better to assembly in a single package two different chips (high power part and signal processing part). Moreover, this separation allows taking benefit of the most advanced technology in each domain (low voltage and high voltage) and it is also easy to update because the two technologies employed can evolve independently. Furthermore, complex control functions can be implemented with an advanced low voltage technology. A typical application that requires managing both low and high voltage/power signal is an off-line Switch Mode Power Supply (see figure 5) [7].

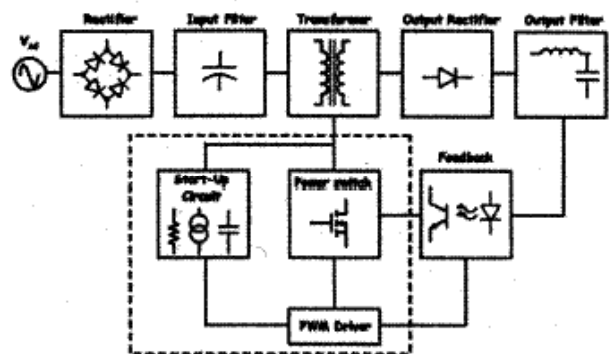


Figure.5. SMPS bloc diagram [7]

For this kind of application, it is possible to integrate in a single package all the elements that are within the dashed box (see figure 5, dashed box), i.e the power switch, the PWM controller and the start-up circuit. Of course, an entirely monolithic solution is also possible, however the cost and performance will be poor compared to the hybrid case (integration in package).

The integrated solution consists of two chips: one chip that contains the power elements and one chip that contains low voltage elements. A bloc diagram that

illustrates the elements integrated in the power chip is given in figure 6. The realisation of this High voltage monolithic IC is achieved using a high voltage like DMOS process standard technology with some integrated feature: current and thermal sensor, start-up system [7]. According to the results reported, the static breakdown voltage of the power device are not affected by the integration. This last solution seems to be well suited to develop new integrated functions for medium power applications and voltage blocking capability in the range 600-1200 V.

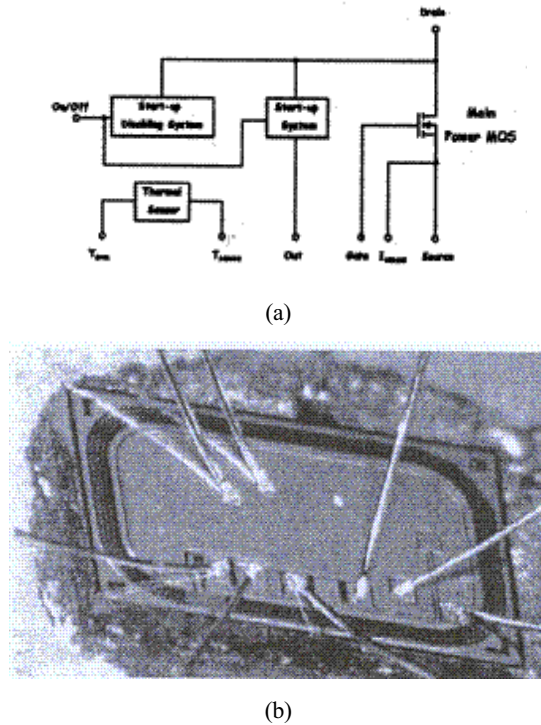


Figure. 6. a) High voltage monolithic IC [7]
b) the realized chip\

DESIRABLE INTEGRATION LEVEL FOR 600-1200V APPLICATIONS

The concept of the functional integration, that allows to devise new monolithically integrated functions using the technology of the power semiconductor device, should allow the integration of all these functions:

1. That require to be close to the power semiconductor device (current, voltage and temperature sensors) [8,9,10,11]
2. Gate driver power supply functions that use high voltage components and hence could share the same termination technique used in the main power device to support the applied voltage. It is to note that integrating this kind of functions with the power device permits a reduction of the system complexity by suppressing all the auxiliary supplies [12].
3. The functions that allow MOS gate control (MOS or LDMOS components capable of supplying high currents)
4. Associated functions (galvanic insulation, micro cooling,...)

Protection circuits integration

Nowadays, power system reliability and ruggedness improvement is one of the power electronics research's concerns. This requires from power semi-conductor devices to be efficient in normal and also in extreme operating modes. Extreme operating modes can be considered as unusual working conditions like high dI/dt and high dV/dt gradients, high dissipated energy and so on. Power devices are usually used in a wide panel of applications. In many of them, they need to be protected from external fault conditions and over temperature conditions are usually implemented in hybrid way with the power device. The natural evolution is thus to integrate in monolithic way the protection circuits into the IGBT in order to increase power system reliability.

- Overvoltage protection: The overvoltage protection circuits allow to clamp the voltage spikes at an acceptable level. Voltage overshoots take place when the device is switched off. An interesting comparison study is carried out on the different integrated solutions in [13]. The various integrated techniques encountered tend to provide an effective voltage clamping while making it possible to benefit from thermal cooling capabilities of power switch. There are basically two large families that can be integrated with the power switch. The first one operates a feedback on power switch gate terminal while the other solution provides a parallel path to dissipate energy. In both cases, however, thermal issues are important to consider since voltage protection circuits may have to dissipate a large amount of heat. For the purpose of illustration, the over voltage protection circuit given in figure 7 consists of a series of poly-crystalline silicon zener diodes and a bulk avalanche diode. This circuit is reported to be monolithically integrated with the IGBT and is able to sustain much higher avalanche energy under the abnormal condition of the voltage overshoot (nearly 400 percent increase in maximum avalanche energy).

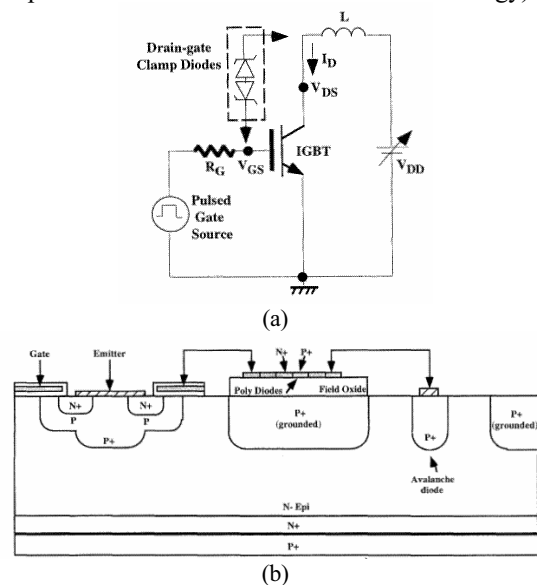


Figure.7. (a) Unclamped inductive switching test circuit (b) cross sectional view of the IGBT with monolithic overvoltage protection circuit [9].

- Over-current, over-voltage and over temperature sensing[14], [15], [8]: an integrated structure reported in literature and represented in figure 8 integrates monolithically all the mentioned protecting and sensing functions. The technology process used for the structure realization is that of the IGBT. The overcurrent sensing and protecting circuits consist of a current sensing IGBT, a lateral n-MOSFET, a poly-crystalline silicon resistor, and a diode. The overvoltage sensing and protecting circuit has a zener diode which is made of the poly-crystalline silicon upon a thick silicon dioxide layer and avalanche diode that is fabricated in the silicon by boron implantation and followed by thermal diffusion. Device temperature was detected by utilizing the temperature dependence of the forward voltage drop of a p/n diode when a specific constant current was forced to flow through the diode. This latter was fabricated using the polycrystalline silicon upon the thick silicon dioxide layer [14].

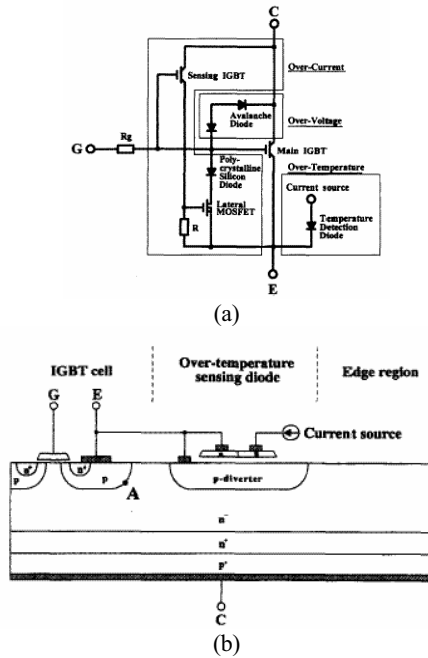


Figure.8. (a) Unclamped inductive switching test circuit (b) cross sectional view of the IGBT with monolithic overvoltage protection circuit [14].

- Short circuit protection: short circuits are one of the main defects that can lead to power device destruction and, unfortunately to power system destruction. Indeed, when a short circuit occurs a simultaneous increase of the voltage and current results in an increase of the dissipated power and consequently in an increase of junction temperature beyond the acceptable level. There exist two types of short circuits: The first type corresponds to the turn-on phase (switching) of the power device and the second type when the power device is in its conducting mode. Various methods for the detection of a short circuit are discussed in [16], [17]. It is, however, important to note that the protection strategy adopted is highly linked to the detection method used: current rise (sensefet), voltage increase across the device (anode voltage sensor). A great

number of works describing some integrated solutions based on senseFET were reported in literature [18, 19]. Recently, monolithically integrated solutions were described in literature and among them one can cite: the structure represented in figure 9(a) that utilizes, to ensure self-protection, a lateral bipolar transistor which is integrated in the main IGBT cell using the 3 dimensional layout [20]. The solution proposed in [21] is given in figure 9(b). It employs an anode voltage sensor that provides an image of the backside anode voltage. The sensor voltage is used as a control voltage of the Mc MOS gate through Md MOS. The anode voltage sensor and the zener diode allows to reach a gate voltage higher than the threshold voltage of the MOS. However, in normal operating conditions, the sensor voltage must be smaller than the threshold voltage of the MOS.

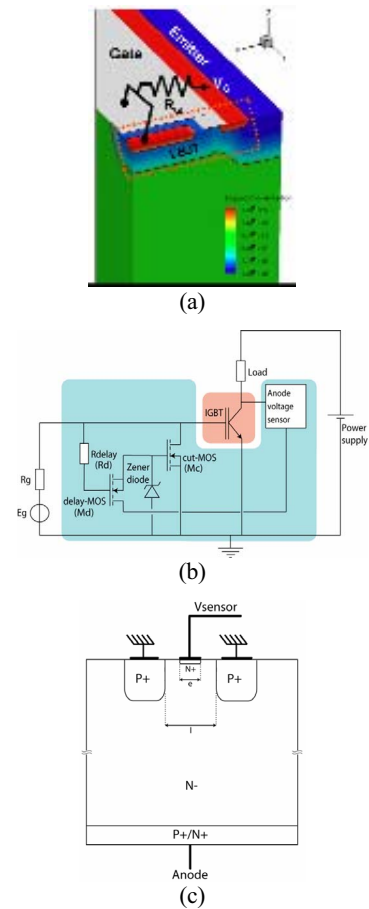


Figure. 9. (a) A self protected IGBT [20] (b)Equivalent circuit of the IGBT short-circuit protection structure [21] (c) and Schematic vertical cross section of the Anode voltage sensor [21]

- Overcurrent equipment protection: to protect circuits or equipments from overcurrents, a solution based on the association of a current limiter [22] in series with the power device was reported. Figure 10 shows a cross sectional-view of the equivalent integrated device where the basic elements can be clearly identified. The basic design procedure of the integrated solution relies on the interconnection of several elementary cells and the whole function is realized using the power device process.

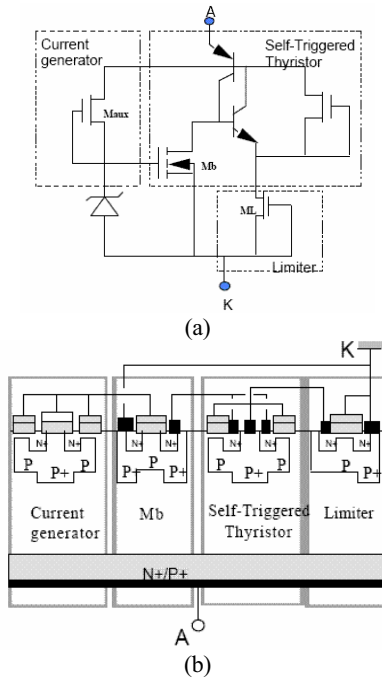


Figure.10. (a) equivalent electrical circuit of the breaker-type protection circuit (b) cross sectional view of the integrated device [22]

MOS gate drive power supply integration

The monolithic integration of a simple, cost effective and reliable gate drive power supply with the power device is essential as far as this would allow replacing the isolated power supplies for gate transistor drivers that are generally a source of volume, cost and failure [23]. Due to the fact that drivers for insulated gate transistors may be referenced to a floating point, their integration is a well known problem. Specific solutions were designed but they are generally implemented outside of the power device [23] and can require complete isolation between components [24]. An attractive solution was proposed in [24] (see figure 11). This gate driver powering technique can be fully integrated (exception made to storage elements, capacitor) within the power device using the power device technological process (see figure 12). Indeed, with this solution, all the power voltage level elements are in parallel and have identical voltage constraints making it possible to share the same junction terminations. The reported technique can be applied both to unilateral insulated gate transistors as well as to bilateral vertical insulated gate transistors [25].

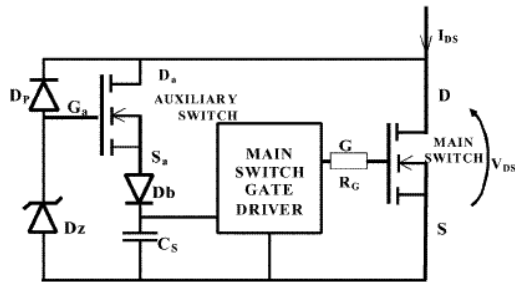


Figure.11. Schematic of a MOSFET switch with its own gate driver and power supply [24]

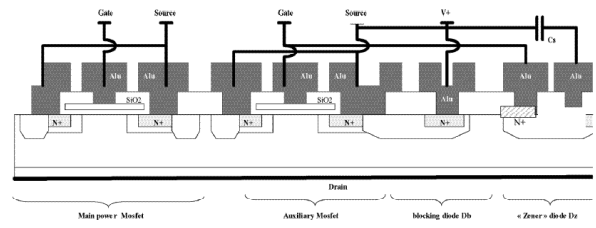


Figure.12. Cross sectional view of the integrated devices [24]

Driver circuit integration

The partial integration of the power switch with its gate driver "IC" will serve to reduce the number of components, decreasing size and improving cost. The "gate driver IC" basically consists of a logic part that controls a powerful driver (high voltage and high current output devices) which provides large output current to charge and discharge the significant gate capacitance of an IGBT. Generally, the monolithic implementation of the output stage (driver) and protection circuits is carried out using a technological process which is different from that of the logic part [26], [27]. An interesting integrated solution is discussed in [28] as shown in figure 13. This method employs a standard low cost CMOS technology to integrate the different parts of the IGBT driver [28]. It is worth pointing out that placing some of the high voltage-high current elements that are located in the driver part with the power switch would make it possible to increase the complexity of the controller (low voltage part) functions that can be realized using the most advanced technology.

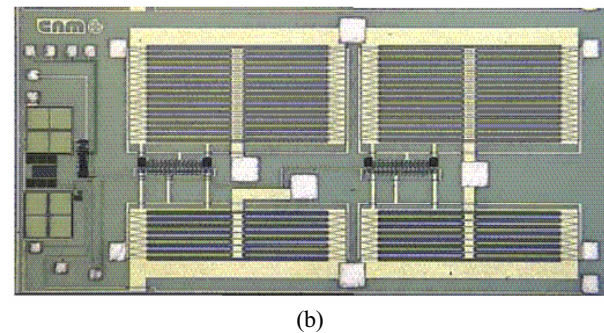
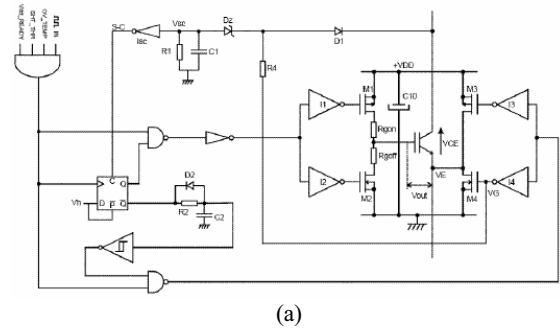


Figure.13. (a) The IGBT gate driver schematic [28]. (b) The realized IGBT gate driver micro-photograph [28].

An analysis of the integration of a low voltage buffer that constitutes the logic part of the chip and which is connected to the gate of the MBS (MOS Bidirectional Switch), given in figure 14, is carried out in [29]. An MBS is an IGBT capable of sustaining reverse voltages. The

logic part is integrated in a P well and therefore it is protected by the reverse biased junction. According to the analysis, of the parasitic coupling that could take place between the power part and the logic part, carried by the authors [29], it appears that there is no significant substrate effect making it possible to reduce the distance between the MBS and the logic part. Furthermore, it is shown that the perturbations could be controlled by adapting the MBS geometry and the output impedance of the logic.

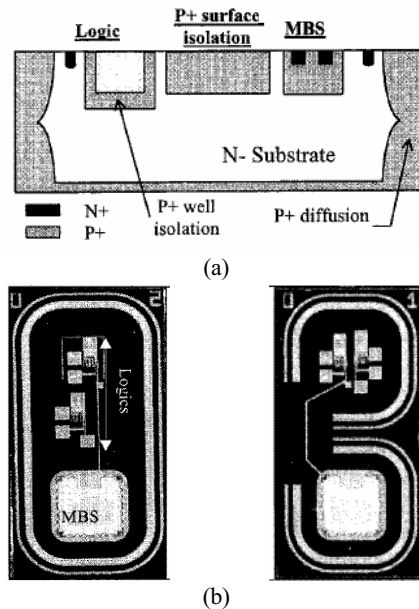


Figure 14. (a) Simplified cross section of the chip [29]
(b) different realized and characterized power ICs [29].

Power device trends for ac applications and implications

Concerning power devices, the development of ac-ac direct conversion circuit or matrix converters requires reverse blocking or full bi-directional power switches (Figure 15 (a)). The symmetrical blocking voltage capability requires specific peripheral junction terminations, and bidirectional current capability requires triggering possibilities on the two sides of the wafer. These requirements can be achieved using P⁺ wall trough wafer, based on thermo-migration of aluminium or in a combination of deep RIE of silicon and deposit of boron doped polysilicon.

Considering bidirectional switches [30,31], an important improvement will be achieved when double sided switches will be available. Besides, if functionalities such as drivers and protections or driver supplies can be integrated, their implementation in ac-ac converter will be greatly simplified. The challenges that one needs to take up in order to realise the bidirectional IGBT as shown in figure 15 (a) are mainly three specific technological steps:

- a- Obtaining a gate oxide of good quality in both sides would be real challenge since generally the gate oxide is realised on one wafer side.
- b- Bonding techniques enabling wafer bonding
- c- Optimisation of specific peripheral terminations that enable bidirectional voltage blocking

As far as driver supplies are concerned, solutions, fully integrated (exception made to the storage elements in a

first step) are under investigation (Figure 15 (b)). Due to their specific behaviour, they are able to operate at high switching frequency and under correct efficiency. Besides, they can be optimized with respect to driver and power switch operating characteristics. An important application of the current/voltage bidirectional IGBT is in a communicating environment. Indeed, in the context of power management in intelligent buildings, the use of a communicating bidirectional switch (obtained by integrating the bidirectional switch and an electronic circuit rendering the ac switch a communicating one) directly connected to the ac mains would contribute on one hand for a better energy management and on the other hand to make the electric installations safer through the use of sensors that could be integrated with the power switch.

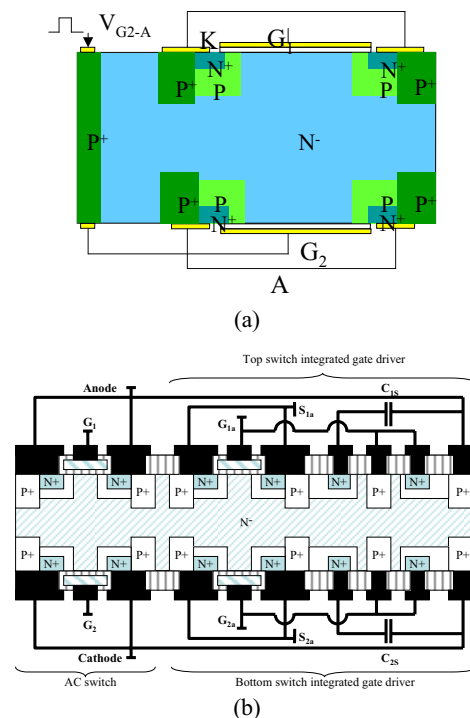


Figure 15. (a) current and voltage bidirectional IGBT
(b) bidirectional IGBT with integrated both side driver supplies [25].

EVOLUTION OF THE TECHNOLOGY

The example of a communicating bidirectional ac switch described briefly above would have to integrate low voltage circuits such as control elements with the bidirectional device. This will certainly require galvanic insulation. To achieve this function (insulation), it is necessary to integrate microtransformers with the power device. Therefore, the variety and the complexity of auxiliary elements that could be integrated with the power device could be schematically represented as shown in figure. 16. The realization will rely on an extensive use of micro-fabrication techniques and specific materials (magnetic, piezoelectric, dielectric, ...). The technological steps for the realization of the elements that are integrated with the power device should be inserted within the power device flexible technological process flow. Deposition, etching and so on gives the opportunity to

operate and to implement in 3D, within the silicon die volume as well as at the silicon die surface.

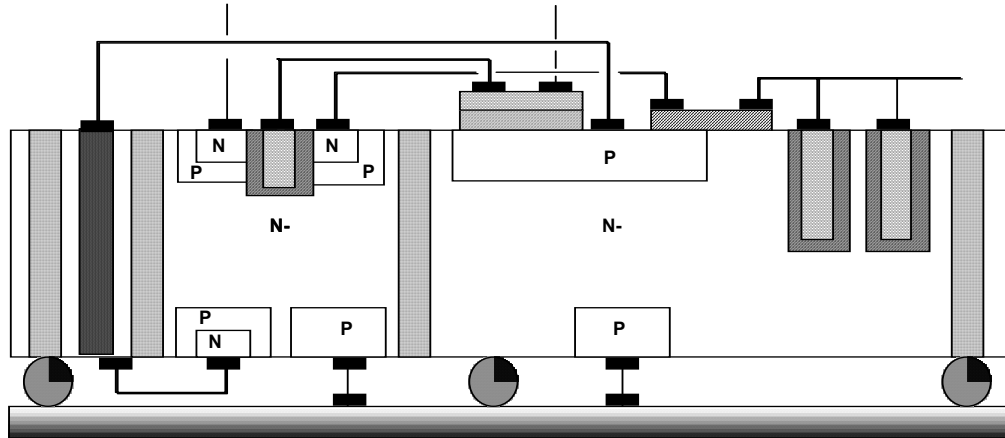


Figure.16. An illustrative cross section of a structure that highlights the trends towards the 3D heterogeneous functional integration.

A flexible technology process

A flexible process must be based on a succession of technological steps corresponding to the fabrication of an IGBT (power device) type structure which constitutes the generic element. The basic steps are defined and optimized so as to include specific steps (see figure 17) supporting more complex functions (control, detection, protection) as discussed in the preceding sections. From these basic elements, other functionalities could be achieved by developing specific technological steps. Preformed channels, for example, are obtained by controlled ionic implantation dose (boron on P type channels and phosphorus of N type) through the oxide and the polysilicon gate. They are

then annealed at a temperature of 950°C. Other building blocks should be integrated on the backside such as N^+ regions making it possible to realize bidirectional devices (antiparallel diodes association with an IGBT) and specific P anodes regions referred to as semi-transparent during fabrication of fast IGBTs. To make the 3D heterogeneous integration “feasible”, the actual flexible technological process must be extended to include microfabrication technology steps that would enable the association of the power device, isolation elements (microtransformers), micro-inductors and silicon capacitors as well as “the on silicon cooling techniques”.

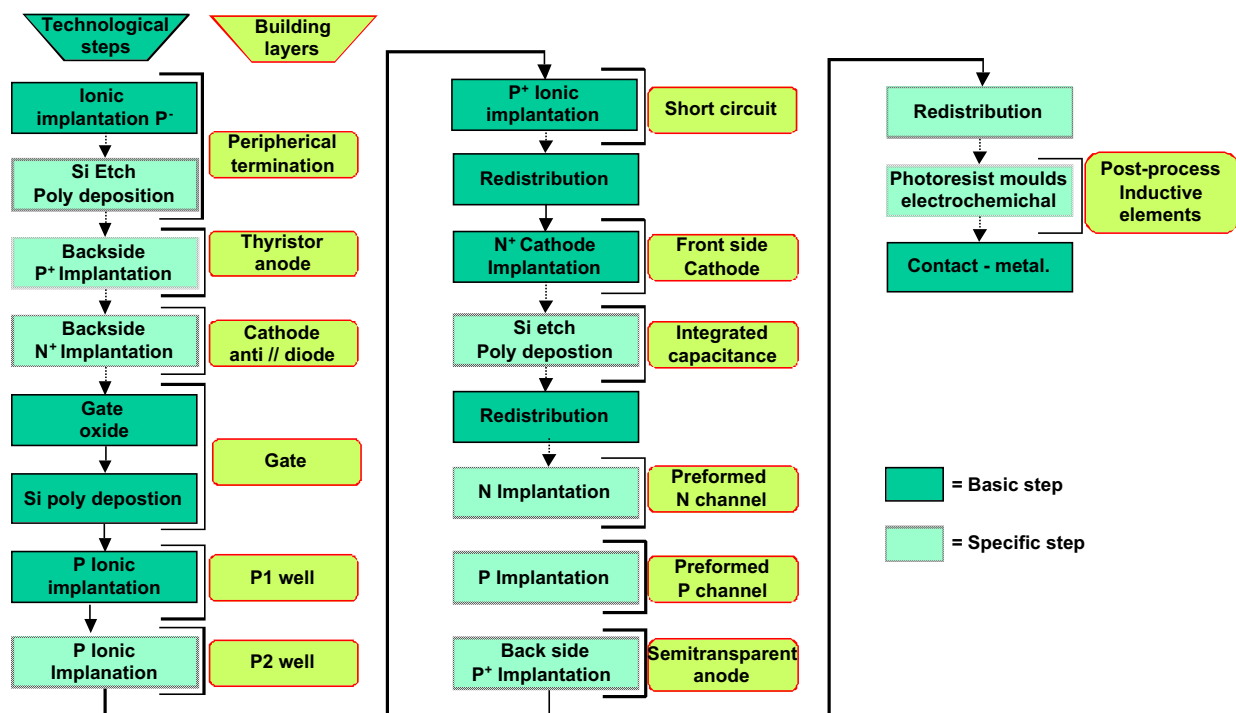


Figure.17. A flexible technological process suitable for 3D heterogeneous functional integration.

Microfabrication techniques for the realization of passive elements

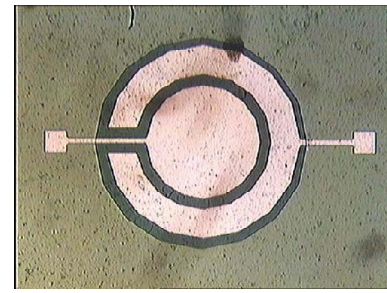
Numerous works carried out in the field of Microsystems have permitted the development of MEMS devices that are mainly used to ensure detection or actuating functions. The fabrication of these MEMS is based on the micro-fabrication techniques.

In the field of mobile electronics, the miniaturisation is also of great importance. Nowadays, the converters size is mainly determined by the sizes of the constituting passive elements (inductors, capacitors, transformers). Consequently, for the above mentioned reasons, the future micro-converters generations impose the integration of passive elements on the silicon wafer or on an insulated substrate. An integrated device permits miniaturisation and a collective fabrication that leads to cost reduction as well as to reliability improvements due mainly to the reduction in the number of interconnections. Many works enabled the realisation of 3D capacitors on silicon as well as inductors. The whole microfabrication techniques could be used today to develop future power integrated functions.

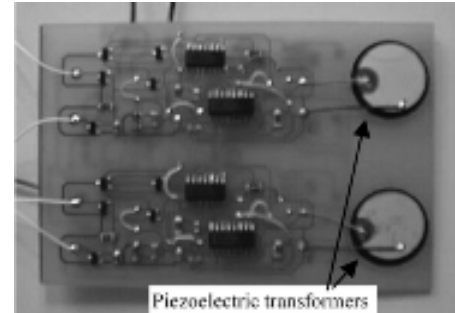
The normal evolution of the classical “functional integration concept” as described previously would be, as highlighted in figure 16, towards the integration of new functionalities (storage, cooling, isolation, ...) giving rise to a **“3D heterogeneous functional integration concept”**. The integration of these functionalities in silicon depend highly on the achievements accomplished in the microfabrication techniques. Indeed, the techniques such as reactive ionic etching (RIE), deep etching, chemical vapour deposition (CVD) low pressure chemical vapour deposition (LPCVD), thick resin deposition, electrochemical deposition, aluminium thermo-migration of aluminium, and new assembly techniques would enable in the future to achieve, isolation between power device and low voltage elements, peripheral termination allowing bidirectional blocking voltage, conducting “via” to enable gate control on the backside of the power device (see figure 15(a)), and also to integrate low voltage elements (as well as polysilicon diodes) on the top surface of power device insulated by the field oxide. For the sake of illustration, few examples on the realizations accomplished in silicon are described hereafter:

Galvanic insulation:

For miniaturisation reasons, the integration of the gate driver of power insulated gate transistors of type MOSFET or IGBT requires the development of integrable galvanic insulating functions. This galvanic insulation can be achieved by a micro-transformer using either a piezoelectric coupling or a magnetic coupling. The integration in silicon of piezoelectric micro-transformers to realize insulation between low voltage and high voltage parts is based on the exploitation of mechanical vibrations of multi-layer membrane structures, a process that is usually employed for the realization of actuators. The mechanism of power transmission in a piezoelectric transformer consists in transforming the electric power into mechanical vibrations (primary armatures) then to recover these mechanical vibrations in the form of electric energy in the second armatures [32, 33].



(a)



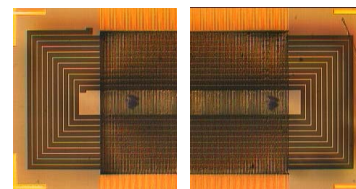
(b)

Figure 18. (a) A microtransformer micrographie [33]
(b) Picture of a gate driver using piezoelectric transformers[33]

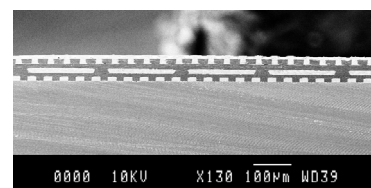
It is however, interesting to envisage the realization of electromagnetic microtransformers on silicon by exploiting the technology breakthroughs achieved in the realization of magnetic core micro-inductors on silicon.

Microinductor and magnetic micro-devices:

Generally, the electro-deposition technology process is adopted to realize magnetic core micro-inductors (figure 19). This process based on the succession of layers electro-deposition (magnetic material as well as copper), however the thickness of the different layers is high (few tenths of microns), hence the layers realisation requires the use of thick photosensitive resists and to make layers stacking possible, a polishing step is mandatory (chemical-mechanical polishing) [34]. The inductance value that can be obtained using this kind of fabrication techniques is about 1 μH . These technologies must be optimised to develop magnetic micro-devices that ensure insulation functions for the control of MOS gated power devices.



(a)



(b)

Figure 19. (a) a micro-inductor micrographie FeNi magnetic core [34]
(b) micrographie of a cross section of micro-inductor FeNi magnetic core [34]

Silicon capacitors:

To reach a high capacitance per unit surface, a compromise on the dielectric thickness is necessary: the minimum dielectric thickness is set by the dielectric breakdown voltage. However, the increase of the surface of the face-to-face electrodes is a purely geometric parameter that can be easily adjusted. For silicon structures, by creating, through etching, trenches or pores in the wafer, the developed surface is increased while the apparent surface is maintained at its minimum. The fabrication of three dimensional capacitors using silicon etching has been demonstrated for decoupling functions. A multi-layer ONO (Oxide-Nitride-silicon Oxide) is deposited by LPCVD (Low Pressure Chemical Vapor Deposition) in the high form factor pores (figure 20). A doped polysilicon is used as a top face electrode. The obtained results are promising since the capacitance per unit surface is about 40 nF/mm^2 [35].

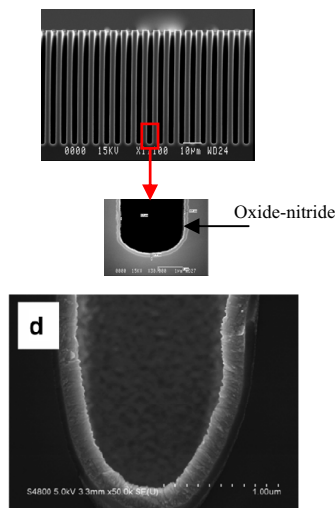


Figure 20. 3D capacitors, deep silicon etching high K deposition in trenches [36]

Micro-Cooling:

In the last ten years, due to the microelectronic fabrication technology progress, the use of silicon for the realisation of micro heat pipes dedicated for cooling of electronic components has received much attention from the scientific community.

The utilisation of silicon micro heat pipes enables to reach thermal conductivities higher than those of copper. Silicon heat pipes based on square or hexagonal columns (figure. 21) can present interesting performances (maximum power of tens of Watts and equivalent thermal conductivity that can reach $800 \text{ W.m}^{-1}.\text{K}^{-1}$ for a heat spreader like utilisation). An experimental study showed that these devices operate for a power of 10 W and in all positions.

Consequently, silicon could even, in the long term, enable the realisation of the electronic component in the same material that would have as a consequence the elimination of the contact thermal resistance as well as the degradations resulting from the use of different materials which expand and contract at different rates during the thermal cycle [37, 38, 39].

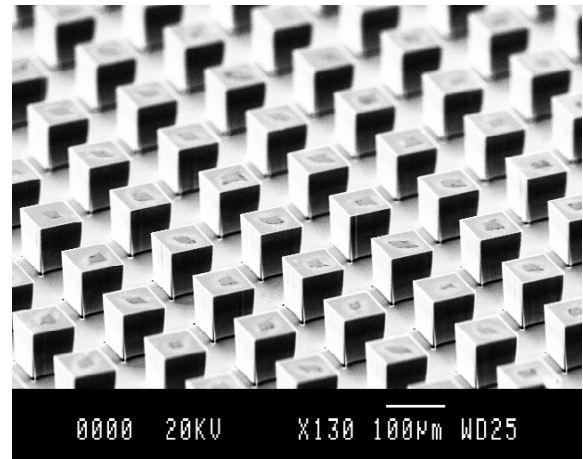


Figure. 21: a micro photograph of a capillary wick structure made in silicon technology [40]

CONCLUSION AND TRENDS

In this paper, a description of some of the realizations achieved using the classical functional integration concept permits to highlight the gradual increase of the number of functions integrated monolithically. The type of integration mode (mixing low power circuits with power processing devices) depends more upon the complexity of the signal processing functions as well as on the amount of power processed by the power elements. Meanwhile, impressive breakthroughs were achieved in the technology for the realisation of passive elements on silicon wafers. Today, the passive elements integration permitted the reduction of low DC power converters size, however, their use could be extended in future for the realisation of new functionalities such as communicating switches as described previously.

As illustrated in figures 22 and 23, a long term objective could be the integration of the switching function (Integrated Power Switch IPS), i.e the power semiconductor device, control functions, protection functions, isolation, storage elements and may be even the cooling mechanism. A direct link between the “functional integration concept” and the “technology evolution” will allow to develop new functionalities thanks the development of new materials (out of the mainstream of the silicon technology currently used for the development of power integrated devices) and specific technological steps. For evident cost reasons, this switching function would be realized by assembling diverse silicon chips using a “collective fabrication process” (wafer level packaging). Consequently, the integrated switch will be a 3D heterogeneous object looking much like some of the currently developed Microsystems.

From the design point of view, it would be interesting to make available for the designers, a design environment and strategy that will guide them all along the design process starting from the imposed specifications of the switching structure and ending at the corresponding monolithically realized structure, that could be realized, for instance, using the flexible technological process described previously. Indeed, the specifications are generally established by system designers with little knowledge of the integration constraints. Within this

context, a set of design rules, simulation software and library models, that take into consideration electrical as well as technological aspects need to be developed in order to set up the bases of an efficient design environment.

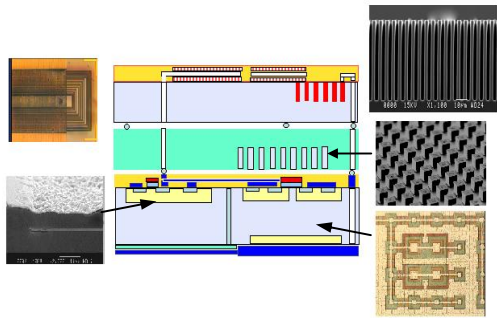


Figure .22. A possible power microsystem according to the 3D heterogeneous functional integration principle.

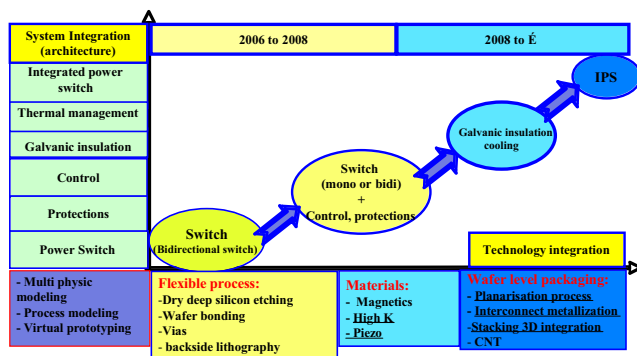


Figure .23. Integrated power switch progression

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